Héctor Gabriel Santana Rosario

Computer Engineering | Robotics | Hardware | Software

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Objective

In my professional development as an engineer, I have three areas of interest: education, growth, and learning. I want to demonstrate my capabilities in multiple areas while also highlighting other's abilities. I want to make it a mix of my interests where I educate myself, while also helping others grow and learn. It is of great importance to have a solid base of knowledge when explaining it to others, since this can lead to immense progress on a project. In general, I look forward to putting to test my technical skills, while also demonstrating my leadership capabilities when working on a project.

Education

Relevant coursework – Electronics I, Electrical Circuits I & II, Physics for Engineers I & II, Sustainable Engineering, Advanced Programming, Software Design and Construct.

Inter American University of Puerto Rico – School of Engineering *August 2024 – present*

- Pursuing a Bachelor of Science in Computer Engineering
  - Expected Graduation Date - May 2028
- GPA: 4.00

Academia Santa Rosa de Lima, Bayamón, Puerto Rico *August 2022 – May 2024*

- 12th Grade Diploma – Member of the Honor Society
- GPA: 4.00

Scholarships

Inter American University of Puerto Rico – Bayamón Campus *August 2024 – present*

- Engineering Scholarship
- Honor's Program Scholarship

AEELA Scholarship *August 2026*

Recognitions

Inter American University of Puerto Rico – Bayamón Campus *2025 – present*

- Member of the Chancellor's List

Co-curricular and Extra-curricular Activities

Inter American University of Puerto Rico Bayamón Campus *2025– present*

- Eco Exploratory
  - As part of the Inter Robotics team, we presented our test robot and its functionalities in which I explained the programming section by demonstrating its capabilities and describing in general how parts of the code work.
- CIAPR Expo
  - Representing the Inter Robotics team, we went to this event and presented our robots for people to see our project's progress. We managed to obtain a prize for creativity.
- The Inter Robotics team expanded for taking on a new project with Raytheon in a drone project, in which I take part as the co-Captain.
- Took part in the national competition for Vex Robotics at Gurabo PR as part of the Inter Robotics team where we obtained the creativity award.
- Competed in the hackathon competition i-HACK.
- Participated in the "Concierto primavera, Voces que inspiran" as part of the group as a pianist on May 20, 2025, held in the university theater.

Projects

2026 (Concurrent)

- VEX Competition: Override
  - For this season, we have to create a robot capable of stacking up pins and cups. Like before, it will consist with a total of two robots, where I am mainly working on the leadership of the programming sub-system.
- Raytheon Autonomous Vehicle Competition AVC
  - We have a mission in particular, to create an unmanned vehicle in which we can communicate through natural language for its movement, guide it to an objective and retrieve a message through NFC, and communicate that message to the pilot. It covers multiple areas in which engineering students of different areas are required.
- Circuit Generator

    - For our Circuits II project, we are creating a program in which the user can select certain components while giving the necessary values in order to output the visual representation of the circuit in which all the possible values will be shown and listed.
- Intelligent Notes
    - For the Software Design & Construct course we are working on developing our technical skills when working on a project. For this, we are creating a multi-platform application for taking notes in which we give the user multiple design and technical options, ensuring efficient work.

2025(Completed)

- VEX Competition: Push Back
    - In this season's project, we had to create a robot for picking up blocks and placing them in a goal. It consisted of two robots in total where I mainly participated on the driver control code.
- Sleep Mind
    - This project was done by my teammates Jaydam and Nancy, and me. It consists of a Python-based program that uses an LLM for analyzing sleeping patterns, taking data of your sleeping habits and outputting a Spotify playlist for improving your next sleeping session.

Student Organizations

2025 – present

- Inter Robotics: Lead Programmer
- Inter Robotics AERIS branch: Co-Captain
- Computing Alliance of Hispanic-Serving Institutions: Member
- Student Chapter of "Instituto de Ingenieros de Computadoras": Vocal
- Student Association of the Institute of Electrical Engineers: Member

Skills

- **Programming & Systems:** C++, Java, Python
- **UI & Systems Development:** QML (Qt Modeling Language), Multisim, Logisim
- **Hardware & Embedded Systems:** Raspberry Pi, SDR (Software-Defined Radio)
- **Diagramming & Technical Documentation:** MERMAID
- **Version Control:** GitHub, GitLab
- **Design & Workspace management:** Notion, Canva
- **Productivity Software:** Microsoft Office (Word, Excel, PowerPoint)
- **Capabilities:** leadership, communicative, problem solving, researcher, responsibility, teamwork, time management.
- **Languages:** English (High Proficiency), Spanish (Native)
- **Creative Pursuits:** Piano